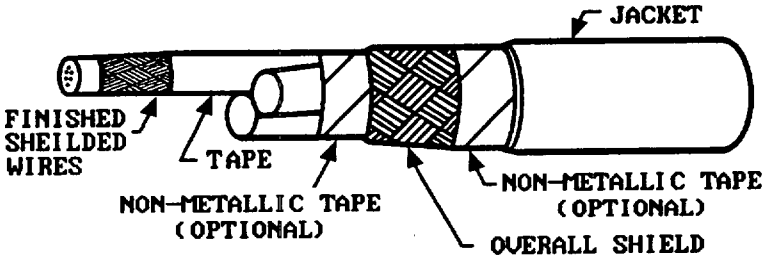
INCH-POUND
MIL-C-24640/14B
31 July 1989
SUPERSEDING
MIL-C-0024640/14A(SH)
23 October 1985
USED IN LIEU OF
MIL-C-24640/14
10 August 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, LIGHTWEIGHT, MULTI-CONDUCTOR, AWG 22, TYPE 1XSOW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.



SH 12761

FINISHED SHIELDED WIRE:

(B) First - Tin coated copper conductor, size AWG 22, 7 stranding in accordance with ASTM B 286. Nominal diameter of stranded conductor 0.031.

Second - Insulation material requirements in accordance with MIL-W-81044/12 except the conductor and identification codes (B) and methods shall be in accordance with MIL-C-24640. Standard identification code applied by method 1. Finished wire diameter 0.044 inch minimum and 0.050 inch maximum.

(B) Third - Shield shall be tin-coated copper braid with a coverage not less than 90 percent determined in accordance with MIL-C-27500.

FIGURE 1. Construction (watertight).

(B) denotes changes.

AMSC N/A FSC 6145

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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(B) TABLE I. Construction details.

PIN M24640/14	Type number	Number of conductors	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
					(min)	(max)	
-01U0	1XSOW-2	2	0.173	0.034	0.292	0.314	83.0
-02U0	1XSOW-14	14	.362	.041	.470	.506	233.0
-03U0	1XSOW-20	20	.429	.044	.542	.584	307.0
-04U0	1XSOW-30	30	.520	.044	.614	.662	429.0

REQUIREMENTS:

Qualification required.
Construction details: See table I.

(B) Finished wire:

For the finished wire the following additional tests shall be applicable in accordance with MIL-W-81044:

WIRE, RATINGS AND ADDITIONAL REQUIREMENTS:

Temperature rating: 135°C (275°F) maximum conductor temperature.
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 250 ± 2°C (482 ± 3.6°F) for 6 hours; identification legibility, 225 ± 2°C (437 ± 3.6°F) for 6 hours.
Blocking: 150 ± 2°C (302 ± 3.6°F).
Color: In accordance with MIL-STD-104, class 1; white preferred.
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Identification, striping, or banding durability: 125 cycles (250 strokes) (min), 500-gram weight.
Impulse dielectric test:
Primary insulation (when test is used in lieu of spark test): 6.0 kilovolts (peak), 100 percent test.
Finished wire: 8.0 kilovolts (peak), 100 percent test.
Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200 ± 2°C (392 ± 3.6°F) for 120 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).

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REQUIREMENTS: - Continued

Shrinkage: 0.125 inch maximum at 300 ± 2°C (572 ± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz, 100 percent test.
Surface resistance: 500 megohms-inches (min), initial and final readings.
Thermal shock: Oven temperature, 150 ± 2°C (302 ± 3.6°F).
Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

(B) Overall shield shall be tin-coated copper braid with a coverage not less than 90 percent determined in accordance with MIL-C-27500.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
(B) Conductor resistance - ohms/1000 feet, at 25°C, maximum	17.63
Voltage withstand - volts, root mean square, minimum	
(B) Conductor to conductor	Not applicable
Conductor to shield	1000
Shield to shield	500
Insulation resistance - megohms/1000 feet, minimum	5000
Conductor and shield continuity	No failure
Jacket flaws	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
(B) Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	120
(B) Capacitance unbalance - percent, maximum	Not applicable
(B) Characteristic impedance - at 1 MHz, ohms	13 min, 27 max
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	No failure
Weight	No failure
Group B:	
Cold bending	No failure
Drip	Zero
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
(B) Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
Gas flame (1 hour)	Not applicable

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EXAMINATION AND TESTS: - Continued

	Requirements
Group C:	
Physicals (aged)	
Jacket	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Armor - conformance to material, construction and coverage	Not applicable
Shield - conformance to material, construction and coverage	No failure
Heat distortion - percent unaged, maximum	30
Cable filler removability	No failure
Shrinkage	No failure
Mutual inductance - dB/1000 feet, maximum	Not applicable
Surface transfer impedance	
Milliohms per meter, maximum	700
EMP response, dB, minimum	60
Pulse response time - ms/500 feet	
Rise time, maximum	Not applicable
Decay time, maximum	Not applicable

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, and C, plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation	No failure
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50

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QUALIFICATION INSPECTION: - Continued

	Requirements
Smoke index, maximum	
Jacket	25
Fillers	--
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer - (type A) hardness, minimum	80
Weathering	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Finished wire	Not applicable

Preparing activity:
Navy - SH
(Project 6145-1004-14)

Custodians:
Army - MI
Navy - SH
Air Force - 85

Review activities:
Navy - EC
Air Force - 99
DLA - IS

User activities:
Army - ME, AR
Navy - CG

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